

Linear Transformations

Vector spaces are only half of the story; the other half is the maps between them that respect addition and scalar multiplication. This chapter studies these *linear transformations*.

3.-1.1 Overview

In this part, we will explore:

- **Linear transformations:** the definition and basic properties, examples and non-examples, and the fact that a linear transformation is determined by its values on a basis.
- **Kernel and image:** the two subspaces attached to every linear transformation, with examples and their basic theorems.
- **The rank–nullity theorem:** nullity and rank, and the identity $\text{nullity}(f) + \text{rank}(f) = \dim(V)$ for a finite-dimensional V .